

Albireo
930 Suns

Sadr
35,250 Suns

Deneb
51,620 Suns

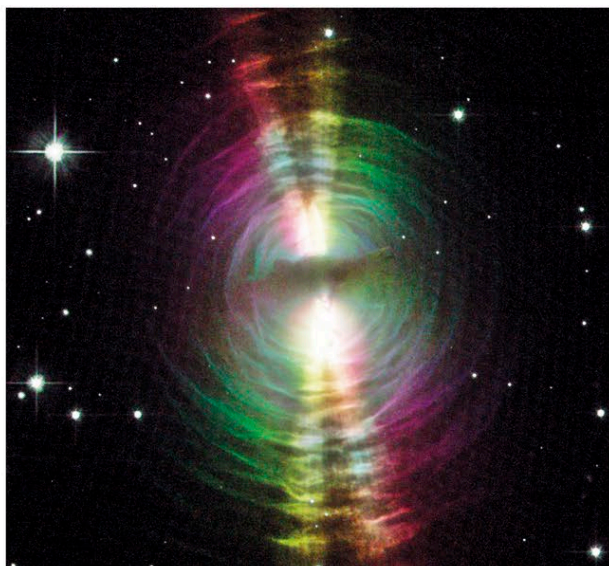
CYGNUS THE SWAN

SOMETIMES REFERRED TO AS THE NORTHERN CROSS BECAUSE OF ITS DISTINCTIVE SHAPE, CYGNUS IS PROMINENT IN NORTHERN SKIES. THE MILKY WAY'S HAZY BAND RUNS THROUGH IT, DIVIDED INTO TWO STREAMS BY THE CYGNUS RIFT—A DARK LANE OF DUST AND GAS.

Cygnus is an ancient Greek constellation representing a swan. Myths tell how the god Zeus turned himself into a swan to pursue the beautiful Queen Leda of Sparta, and the constellation commemorates his disguise. Deneb, the brightest star in Cygnus, lies in the swan's tail, while the bird's long neck extends along the Milky Way to the star Albireo, a true binary star (see pp.40–41) that marks its beak. Other stars suggest the swan's outstretched wings.

Near Deneb lies a cloud of glowing gas named the North America Nebula, since it looks very much like that continent in shape. Difficult to see with smaller instruments, the nebula shows up best on long-exposure photographs.

Between Epsilon Cygni and the border with Vulpecula lies NGC 6992, another nebula best seen on photographs. Known as the Cygnus Loop or Veil Nebula, it is the remains of a star that exploded some 5,000 years ago.



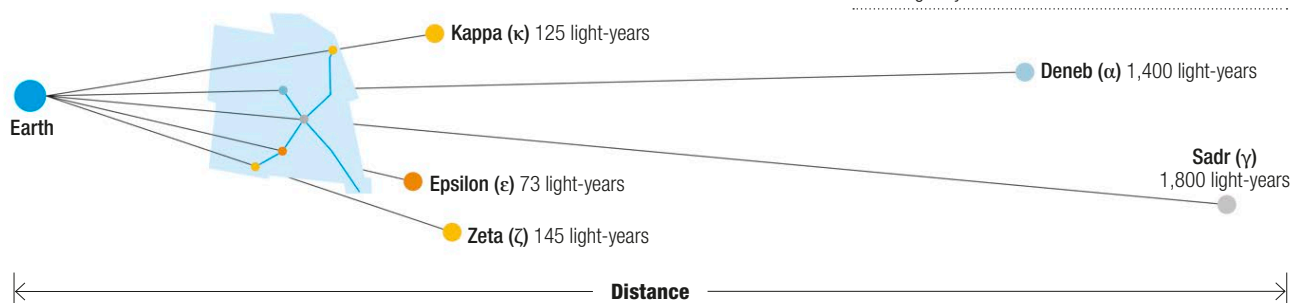
◁ **Egg Nebula**
Starlight shining through thin shells of dust creates beautiful patterns in this color-enhanced Hubble Space Telescope image of the planetary nebula. A thicker inner dust belt blocks out light from the central star.

△ **Cygnus A**
One of the strongest radio sources in the sky, this galaxy has a central supermassive black hole, from which jets of gas (colored red) are thrown out. Hot X-ray-emitting gas is shown in blue.

About **6,000 light-years** from Earth, Cygnus X-1 is a **black hole** that orbits a **blue supergiant** every 5.6 days

► Star distances

The constellation of Cygnus is the 16th-largest constellation and spans vast distances in space when the distances to its pattern stars are taken into account. The tail (Deneb) of the Swan is over 1,000 light-years from its beak (Albireo). Sadr, in its chest, is further away still.



KEY DATA

Size ranking 16
Brightest stars Deneb (α)
1.25, Sadr (γ) 2.2
Genitive Cygni
Abbreviation Cyg
Highest in sky at 10pm
August–September
Fully visible 58°N–83°S



CHART 4

MAIN STARS

Deneb Alpha (α) Cygni
Blue-white supergiant; brightest star in Cygnus
☼ 1.25 ↔ 1,400 light-years
Albireo Beta (β) Cygni
Wide double star; colours are orange and blue-green
☼ 3.1, 5.1 ↔ 400 light-years
Sadr Gamma (γ) Cygni
White supergiant in the middle of the northern cross
☼ 2.2 ↔ 1,800 light-years
Delta (δ) Cygni
Binary star; period 920 years
☼ 2.8 ↔ 165 light-years
Gienah Epsilon (ε) Cygni
Orange giant
☼ 2.5 ↔ 73 light-years
Zeta (ζ) Cygni
Yellow giant; spectroscopic binary
☼ 3.2 ↔ 145 light-years
Mu (μ) Cygni
Binary star; period 790 years
☼ 4.5 ↔ 720 light-years

DEEP-SKY OBJECTS

M39
Open cluster of around 30 stars
NGC 6826 (Blinking Planetary)
Planetary nebula
NGC 6992 (Cygnus Loop / Veil Nebula)
Supernova remnant
NGC 7000 (North America Nebula)
Emission nebula
Cygnus X-1
X-ray binary system containing a black hole
Egg Nebula
Planetary nebula
Cygnus A
Radio galaxy